

Probability reminders



- Three common ways to describe probability:
 - ▶ probability distribution
 - ▶ probability density
 - ▶ measure

Distribution

- *Idea: assign numbers to atomic events so that, if we assign (say) 0.23, the event happens 23% of the time*
- A probability distribution function p maps discrete atomic events to real numbers, with
 - ▶ $p(a) \geq 0$ for all atomic events a
 - ▶ $\sum_{a \in U} p(a) = 1$ summed over universe U (entire table)
- E.g., uniform on n atoms: $p(a) = \frac{1}{n}$ for each
- Notation: $p_X, p_Y, p_{X,Z}$ — but often skip subscripts

$P(X, Y, Z)$

$Z \rightarrow$ $X, Y \downarrow$	0	1
00	0.1	0.2
01	0.2	0.0
10	0.1	0.3
11	0.1	0.0

also called pmf

atomic event = smallest
division of outcomes
e.g.,  = 2, not
 $\in \{\text{even \#s}\}$

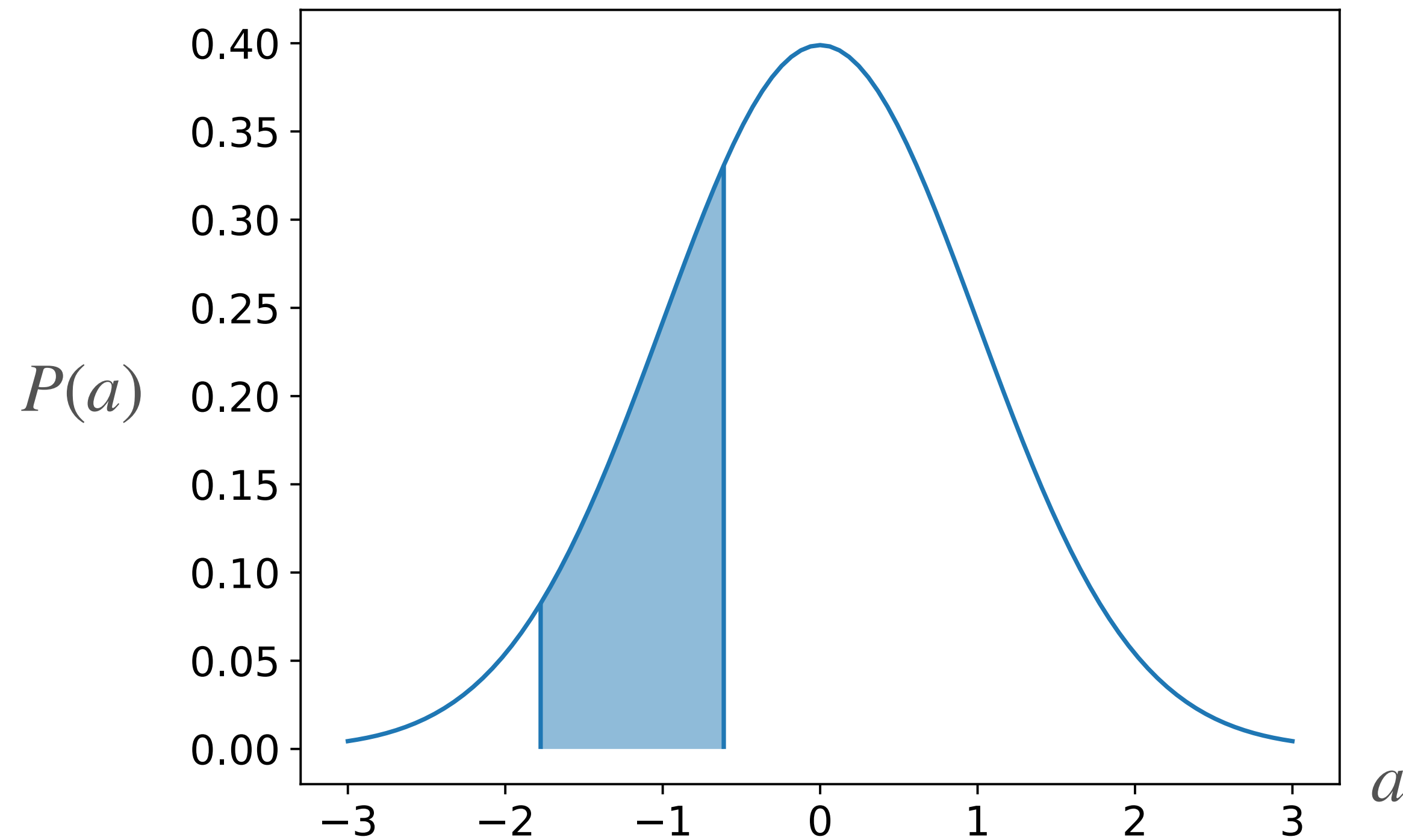
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Density

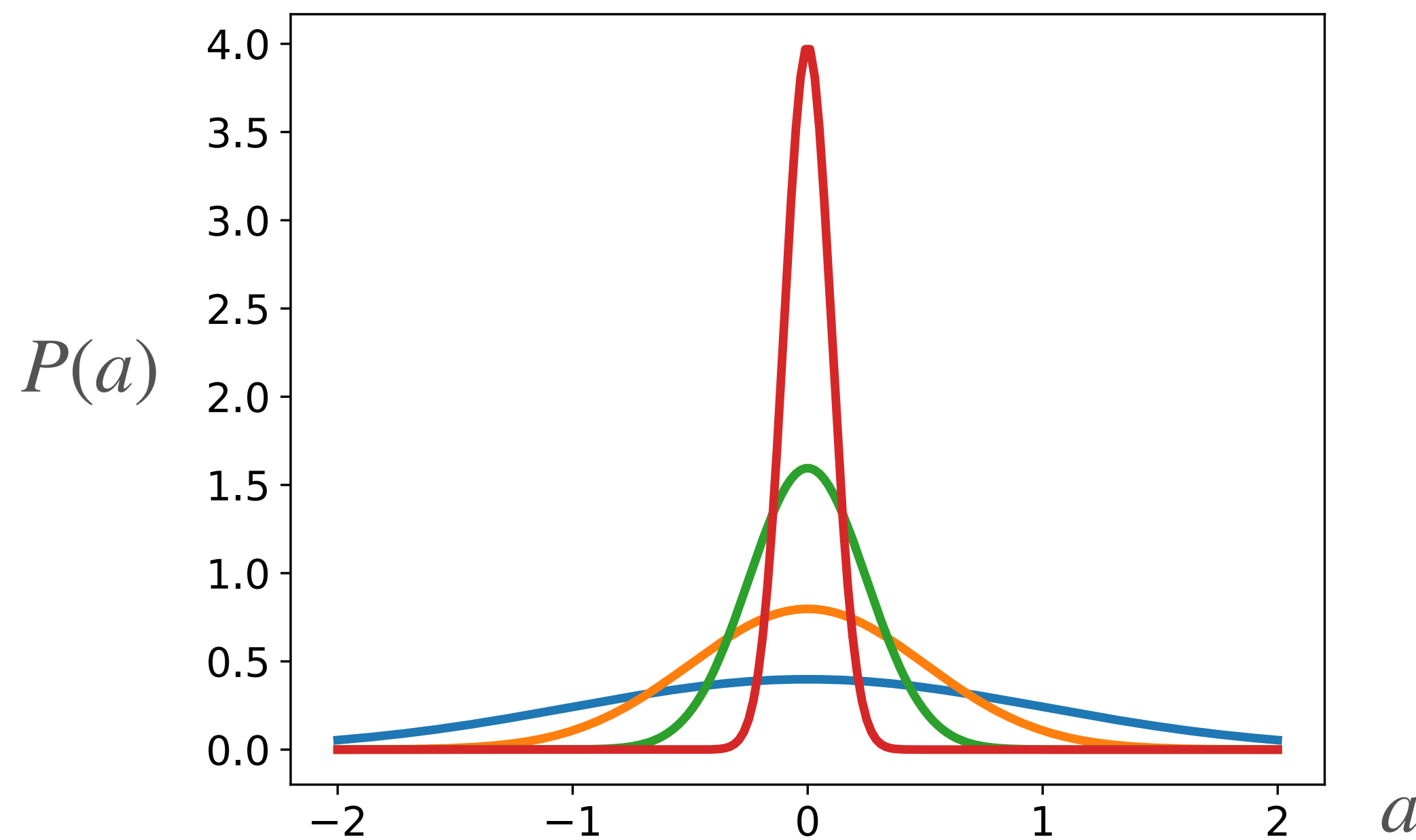


- *Idea: assign numbers to atomic events, so that integrating over atomic events yields a probability*
- A probability density function p maps continuous atomic events to real numbers, with
 - ▶ $p(a) \geq 0$ for all atomic events a
 - ▶ $\int_{a \in U} p(a) da = 1$ over universe U

PDF

- Some statisticians thought they were clever making PDF be the acronym for both
 - ▶ and we use $p(\cdot)$ or $P(\cdot)$ for either
 - ▶ and say distribution even if we mean density (or measure)
 - ▶ worse, can see different uses in the same equation (!)
- Makes some sense since a lot of the same rules happen to apply to both representations
- But they are *not the same thing!*

***Not the
same thing***



- E.g., for a density, $p(0.37) \neq$ probability of seeing 0.37
 - ▶ in fact, probability of any atomic event = 0!
- E.g., for a discrete distribution, $p(a) \leq 1$ for each outcome (atomic event) a , but for a continuous density, $p(a)$ can be arbitrarily large
- Density acts like the *derivative* of probability

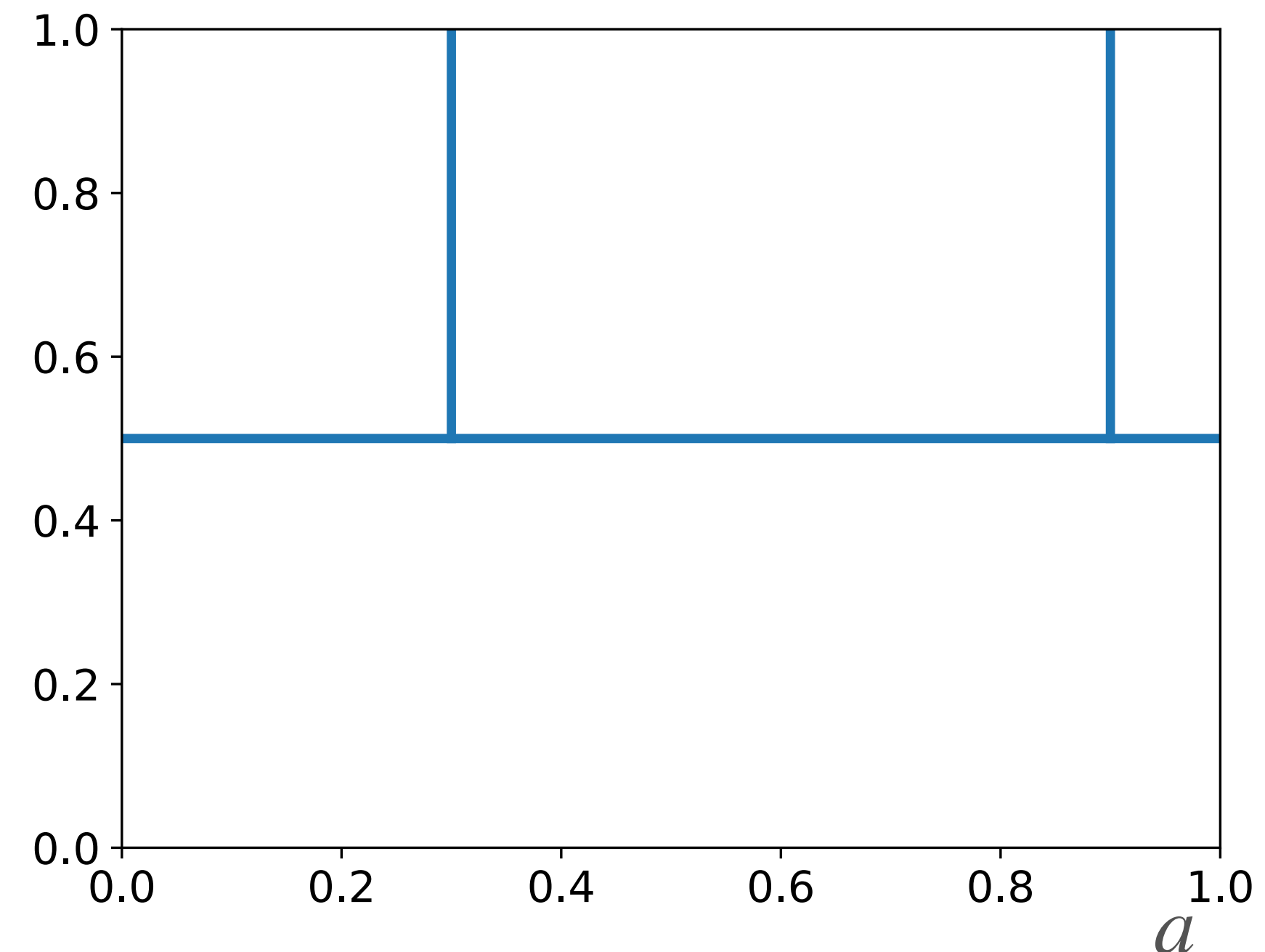
$$p(E) = \int_{a \in E} p(a) da \text{ for an event } E$$



Measure

won't typically use this
in 10-701

$P(a)$



- *Idea: directly define the result of integration, enforcing only rules of nonnegativity, subadditivity, and $P(U) = 1$*
- Generalizes both discrete and continuous PDFs
- Notation is like continuous: $p(E) = \int_{a \in E} \mu(a) da$
- But OK to have atomic events with $p(a) > 0$
 - ▶ imagine a Gaussian bump that's infinitely narrow centered at a (for physicists among us: a Dirac delta)
 - ▶ e.g., $p(x) = 0.5 + 0.25 \delta(x - 0.3) + 0.25 \delta(x - 0.9)$
 $x \in [0, 1]$

Conditional probability

- Describe w/ *conditional* distribution, density, or measure
 - ▶ same as above, except $p(\cdot)$ takes conditioning event as another input, $p(X, Y \mid Z = z)$
 - ▶ sums or integrates to 1 over X, Y for *each* atomic event z (but not other way)
- Can think of $p(X, Y \mid Z = z)$ as a function of just x, y (with z fixed and compiled in), or as a function of both x, y and z
 - ▶ to emphasize latter, write $p(X, Y \mid Z)$

$P(X, Y \mid Z)$		
$Z \rightarrow$ $X, Y \downarrow$	0	1
00	0.1	0.5
01	0.2	0.3
10	0.6	0.2
11	0.1	0.0

Factorization

$P(X)$	$P(Z \mid X)$		
$\frac{1}{3}$	$\frac{1}{2}$	$\frac{1}{2}$	$\uparrow$ X
$\frac{2}{3}$	$\frac{2}{3}$	$\frac{1}{3}$	
	$Z \rightarrow$		

- Given joint distribution (or density or measure) $P(X, Y, Z)$
- We can always factor it in any order:
 - ▶ $P(X \mid Y, Z) P(Y \mid Z) P(Z)$ [order X, Y, Z]
 - ▶ $P(Y \mid X, Z) P(Z \mid X) P(X)$ [order Y, Z, X]
 - ▶ ...
- How to interpret the above expressions?
 - ▶ discrete case: each term such as $P(Y \mid Z)$ is a table (# outcomes for $Y \times$ # outcomes for Z)
 - ▶ continuous case: each term such as $P(Y \mid Z)$ is a function (arguments Y, Z — continuous analog of a table)
 - ▶ product = *broadcast* and *multiply corresponding* entries

Bayes rule

- By factorization:
 - ▶ $p(X) p(Y | X) = p(X, Y) = p(Y) p(X | Y)$
- Divide through by $p(X)$:
 - ▶ $p(Y | X) = p(X | Y) p(Y) / p(X) \quad \leftarrow \text{Bayes rule}$